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SUPERSEDING
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9 August 1955

MILITARY SPECIFICATION

ENGINE; ROCKET, SOLID PROPELLANT 5 SECOND DURATION, 4500 POUNDS THRUST

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification established the requirements for a 5 second duration, 4500 pound thrust (when determined at 60°F) solid propellant rocket engine hereinafter specified as 5 S-4500.

1.2 Classification. The 5 S-4500 rocket engine is a moderate duration, low-thrust, self-contained power plant employing a solid propellant. The engine is designed specifically for jet assist take-off (JATO) of aircraft, and to deliver a nominal thrust of 4500 pounds for 5 seconds duration at 60°F and sea level. The engine contains aircraft attachment fittings and an ignition system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2845

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
DoD-D-1000	Drawings, Engineering and Associated List
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-D-5480	Data, Engineering and Technical, Reproduction Requirements for
MIL-I-6867	Magnetic Inspection Units
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-I-6870	Inspection Program Requirements, Non-Destructive for Aircraft and Missile Materials and Parts
MIL-F-7179	Finishes and Coatings, Protection of Aerospace Weapon Systems, Structures and Parts, General Specification For
MIL-J-8667	Jet Assisted Take-Off Systems - Design and Installation of, in Naval Aircraft
MIL-C-16173	Corrosion Prevention Compound, Solvent Cutback, Cold Application
MIL-S-28786	Switches, Preparation For Delivery Of

STANDARDS

MILITARY

DoD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-320	Fuse Explosive Component Terminology, Dimensions and Materials
MIL-STD-410	Non-Destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic and Ultrasonic)
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-481	Configuration Control - Engineering Changes, Deviations and Waivers (Short Form)
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders

Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

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2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specifications to the extent specified herein.

DoD Manual

DoD 4145.26M	Safety Manual for Ammunition, Explosives and Related Dangerous Materials
DoD 5154.4S	DoD Ammunition and Explosives Safety Standards

Naval Air Systems Drawings

NAF Dwg. 502706	JATO Unit, 30° Canted Nozzle, 4500 lb. Thrust Envelope Dimensions

Naval Sea Publications

NAVSEA S9086-AA-STM-000	Naval Ship's Technical Manual Chapter 700
OP-5	Ammunition and Explosives Ashore
FEDERAL TEST METHOD 151	Test Methods Metals

2.1.3 Order of precedence. When the requirements of the contract, this military specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

2.1.3.1 Contract. The contract shall have precedence over any specification.

2.1.3.2 Military Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification, or from subsidiary specifications where applicable, shall be specifically approved by the procuring agency.

2.1.3.3 Model Specification. A specification shall be prepared by the contractor in which the particular characteristics of his rocket engine model shall be defined. The model specification shall contain information which is supplementary to the military specification and shall state specifically any deviations from the requirements of the latter. The model specification shall be incorporated as a part of the procurement contract.

3. REQUIREMENTS

3.1 Engine performance characteristics. The performance of the rocket engine shall be reliable and uniform throughout the storage and operating limits specified herein. Performance curves shown in the attached figures shall constitute the performance part of this specification. Variation in average thrust, thrust duration and total impulse shall be kept at a minimum by the best methods of quality and process control. The performance ratings of the rocket engine shall be as shown in Figures 1, 2 and 3.

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3.1.1 Operating temperature range. The engine performance shall be reliable and consistent throughout the temperature range of -30°F to +140°F. To demonstrate this performance requirement, test firings shall be made at -40°F and +150°F, respectively.

3.1.2 Duration of thrust variation. Variation in duration of thrust of the rocket engine fired at any temperature within the operating temperature shall not exceed that specified in Figure 1.

3.1.2.1 Thrust duration measurement. Thrust duration shall be measured on the thrust-time curve (see Figure 4, sample) between the point at which the tangent to the initial rise of the curve crosses the zero thrust ordinate and the point at which the tangent to the final decline of the curve crosses the zero thrust ordinate.

3.1.3 Average thrust variation. Variation of the average thrust of the rocket engine fired at sea level and at any temperature within the firing temperature range shall not exceed that specified in Figure 2.

3.1.3.1 Average thrust. Average thrust shall be obtained by dividing the total impulse (3.1.3.2) by the duration of thrust (3.1.2.1).

3.1.3.2 Total impulse. Total impulse shall be determined by integration of the thrust-time curve from the point where a line tangent to the initial rise of the thrust curve intersects the line of zero thrust, to the point where a line tangent to the final decline of the thrust curve intersects the line of zero thrust.

3.1.4 Total impulse variation. Variations in total impulse (3.1.3.2) of the rocket engine within the operating temperature range of -30°F to +140°F shall not exceed that specified in Figure 3.

3.1.5 Ignition interval limit. The ignition interval at any temperature within the operating temperature range (3.1.1) shall not exceed 0.230 seconds.

3.1.5.1 Ignition interval. Ignition interval is defined as the time from switch operation to the time when a minimum continuous thrust of 3000 lb. at -40°F, 3500 lb. at 60°F, and 4500 lb. at +150°F is obtained during the initial three seconds of thrust duration.

3.2 Qualification Inspection. When specified, a sample shall be subjected to qualification inspection (see 4.2).

3.3 Material. Material used shall conform to applicable government specifications or contractor's specifications. Materials conforming only to contractor's specifications may be used to provided the specifications are approved by the cognizant customer representative. The use of contractor's specifications will not constitute waiver of customer inspection.

3.4 Parts.

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3.4.1 Standard parts. Army-Navy Standard parts and references to their AN part numbers shall be used whenever possible.

3.4.2 Dissimiliar metals. Contact between dissimiliar metals, especially brass, copper or steel, in contact with aluminum or its alloys shall be insulated in the best practicable manner. Refer to MIL-STD-889 for the compatibility for the metals.

3.4.3 Interchangeability. Design of components shall be such as to insure interchangeability of parts and assemblies wherever possible. Quality control of components shall be exercised to assure such interchangeability. Exceptions to this specifications shall be limited to the following:

(a) Nozzle throat diameters shall be assigned specifically for a batch of propellant when throat resizing is required to bring performance within allowable limits.

(b) Safety diaphragms shall be assembled per applicable drawings and specifications, and shall be interchangeable as assemblies only.

3.5 Design and Construction. All drawings shall be prepared in accordance with DoD-D-1000 (Level Established by Contracting Officer) and MIL-D-5480; and shall be reviewed and approved by the customer or its duly authorized representative before being issued for production purposes. Drawing changes shall be made in accordance with the procedure outlined in MIL-STD-480 and MIL-STD-481.

A complete stress analysis for the rocket engine design shall be prepared and submitted to the customer for approval before manufacturing begins.

3.5.1 Engine. The rocket engine shall be composed of the following major components which shall be specified in detail in the applicable Model Specification.

Chamber Assembly
  Attachment fittings
Aft Cap Assembly including:
  Nozzle and weather seal
  Safety release diaphragm
Igniter Assembly
Propellant Grain Assembly
Shipping Cap - Fore End

3.5.2 Chamber assembly. The chamber, when fitted with igniter assembly and aft cap assembly (including nozzle and safety diaphragm assemblies), constitutes a pressure vessel, which shall be constructed in accordance with the best engineering practice to protect the enclosed propellant grain from all foreign matter including moisture and to satisfy operating pressure requirements.

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3.5.2.1. Fittings. The attachment fittings shall satisfy installation under MIL-J-8667. The over-all envelope shall be within the outline established by NAF Drawing 502706, except that envelope deviations not violating the satisfaction of attachment fittings shall be permitted by contractual agreement between customer and contractor.

3.5.2.2 Proof test. Hydrostatic proof test of each pressure vessel assembly as composed of aft cap assembly, chamber and nozzle shall be made of sufficient magnitude to demonstrate that, at the temperature to be encountered in these parts, adequate strength has been provided for the highest predictable operating pressure and the estimated acceleration loads during JATO operation as set forth in the stress analysis (3.5).

3.5.2.3 Welding. All welding on the chamber shall be done by operators certified in accordance with MIL-T-5021.

3.5.3 Aft Cap Assembly. The aft cap assembly shall form the aft and closure of the chamber and retain the propellant grain in position within the chamber. The aft cap assembly shall incorporate the nozzle and safety diaphragm sub-assembly.

3.5.3.1 Nozzle assembly. The nozzle assembly shall be designed for the intended application and pressures.

3.5.3.2 Safety diagram assembly. The safety diaphragm assembly shall be incorporated into the nozzle assembly to afford protection to the chamber against excessive pressure and shall be so designed that if actuated, pieces of the diaphragm are not ejected.

3.5.4 Igniter assembly. A suitable igniter assembly shall be provided which can be shipped separately and easily inserted into the forward boss of the chamber assembly. The igniter assembly shall be capable of producing proper ignition of the main propellant charge, and shall be equipped with a glow plug, containing dual filaments in parallel, which will set off the igniter charge when a minimum current of 10 amperes is applied. The electrical circuit shall consist of a suitable single lead wire, with a cable type protective covering, from the glow plug, and shall terminate with a standard electrical connector for attachment to the aircraft firing circuit. The ground connection shall be provided through the unit and its aircraft attachment fittings.

3.5.4.1 Ignition. Ignition shall in no case take place with 0.04 (or less) amperes current flowing through the igniter circuit for any length of time.

3.5.4.2 Wiring. Wiring on and around terminals shall be protected from moisture and spray to prevent corrosion and shorting of circuits. The igniter shall be protected from moisture as a detail sub-assembly, when packaged in multiple units for shipment and when assembled in the complete rocket engine.

3.5.5 Propellant grain assembly. The propellant grains shall consist of cylindrical masses, suitably restricted or inhibited to burning on surfaces as may be necessary, with provision for the proper support and securing of the propellant grains within the assembled units.

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3.5.5.1 Stability. The propellant shall be dimensionally stable in all orientations of the motor for the times and temperatures specified under cycling performance.

3.5.5.2 Temperature. The autoignition temperature of the propellant shall be not less than 425°F.

3.5.5.3 Combustion products. The propellant combustion products shall be essentially smokeless at relative humidity of the surrounding atmosphere of less than 80%.

3.5.5.4 Propellant resistance. The propellant shall be resistant to detonation by shock, heat, friction, and spontaneous combustion.

3.5.5.5 Fragments. Fragments of a nature that would be harmful to equipment or personnel shall not be emitted during firing of the rocket engine.

3.6 Shipping cap-fore end. The rocket engine shall contain a shipping cap-fore end which will provide protection from the envirnoment during shipping and storage.

3.7 Environmental requirements. The environment which the rocket engine must be capable of withstanding shall be considered in three general groupings, as follows:

3.7.1 Pre-operational environment. The complete rocket engine including propellant charge shall withstand the following environmental conditions such as prevail during shipping, storage, and handling prior to firing. There shall be no degradation that would reduce the reliability or consistency of performance as specified in Paragraph 3.1. The shelf life under any or all of these conditions separately or simultaneously shall be not less than two (2) years. The igniter and rocket engine shall be stored in accordance with Ordnance Pamphlet OP-5 and DoD 4145.26M Safety Manual.

3.7.1.1 Temperature cycling. The engine shall withstand cycling between the extreme temperatures of the operating temperature range (3.1.1). A cycle is defined as a change from one environmental temperature to another, and back to the original.

3.7.1.2 Salt fog. The engine shall withstand exposure to a fog composed of a salt solution for a period of 48 hours as specified in MIL-STD-810, Method 509.1, Procedure I.

3.7.1.3 Dust. The engine shall withstand exposure to an atmosphere of fine sand (sharp edged dust) for the periods totaling 28 hours as specified in MIL-STD-810, Method 510.1, Procedure I.

3.7.1.4 Humidity. The engine shall withstand exposure to repeated humidity and condensation as obtained in cycling as specified in MIL-STD-810, Method 507.1, Procedure V.

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3.7.1.5 Handling. The rocket engine shall be designed to withstand the following independent loadings:

12 g acceleration, normal to the axis
30 g acceleration, parallel to the axis

3.7.1.6 Vibration. The engine shall withstand 24 hours of continuous vibration of a vector amplitude equivalent to 1 g acceleration at the major resonant frequency in the range of 10 to 50 cycles per second and 24 hours of continuous vibration of a vector amplitude equivalent to 1 g acceleration at the major resonant frequency in the range of 50 to 500 cycles per second. Vibration may be in a plane either parallel or normal to the axis of the rocket motor.

3.7.2 Operational environment. The rocket engine shall perform reliably and consistently in accordance with the performance specified under Paragraph 3.1 under any or all of the following environmental conditions taken separately or simultaneously.

3.7.2.1 Temperature range. The engine shall operate in accordance with the specified performance (3.1) when fully conditioned at any temperature within the operating temperature rance (3.1.1) or partially conditioned between two temperatures both in this range.

3.7.2.2 Humidity. The engine shall operate in accordance with the specified performance (3.1) at any condition within the range of pre-operational humidity environment (3.7.1.4).

3.7.2.3 Acceleration. During operation the rocket engine shall withstand the following accelerations acting independently or collectively:

Lateral - (in any plane) 12 g
Longitudinal - (forward or aft) 30 g

3.7.3 Post-operational environment. The expended rocket engine shall be capable of being separated from the aircraft through an ejection system as specified by the customer.

3.8 Protective finishes. Protective treatments, coatings and paint finishes shall be used on all non-working surfaces not corrosion resistant or otherwise suitably protected, in accordance with MIL-F-7179 and MIL-S-5002, as specified herein. During fabrication, metal parts shall be protected in accordance with MIL-C-16173.

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3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-129. The identification data applied to the rocket engine data stencil or decalcomania shall be as follows:

FSN _____
Engine, Rocket, Solid Propellant
Government Model Designation _____
Serial Number _____
Manufacturer's Name or Trademark _____
Contract No. _____
Propellant Batch No. _____
Manufacturer's Part No. _____
Date of Manufacture _____
Storage Temperature -30°F to +140°F
Firing Temperature -30°F to +140°F

3.9.1 Components. Components which may be shipped separately shall be clearly marked as follows:

FSN _____
Name of Component _____
Manufacturer's Part No. _____
Serial No. _____ (if applicable)
Lot No. _____ (if applicable)
Manufacturer's Name or Trademark _____
Contract No. _____

3.10 Workmanship. The workmanship shall be of a sufficiently high quality to insure satisfactory operation, reliability, and durability consistent with the service life and application of the rocket engine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection procedure manual. An inspection procedure manual shall be prepared, shall be approved by the cognizant inspection agency, and shall govern the manufacturer's inspection of all assemblies and components. The quality control procedure established for propellant raw material, processing, and testing shall be included therein.

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4.1.1.1 Quality control records. Quality control records shall be submitted at the time material is presented for acceptance, shall be maintained by the manufacturer for at least one year after delivery of units, and shall be available for review by authorized representatives of the customer.

4.1.2 Inspection of work in progress and completed parts. Rocket engines and all parts and materials entering into the construction thereof shall be subject to inspection during the course of and upon completion of manufacture by authorized customer representatives, in accordance with the approved inspection procedure manual (4.1.1.).

4.1.2.1 Sampling plans. The responsibility for maintaining a complete and adequate inspection system shall rest with the manufacturer, and this system shall be subject to the approval of the customer. All attribute sampling procedures shall be in accordance with MIL-STD-105.

4.1.3 Cognizance of testing. All tests specified under the inspection procedure shall be conducted by the contractor under the cognizance of a duly authorized customer representative.

4.1.4 Acceptance stamp. Each rocket engine, completed igniter unit when packaged separately, and any other component of the rocket engine that is stored, shipped or packaged separately, shall be stamped with the manufacturer's acceptance stamp and the customer's acceptance stamp.

4.1.4.1 Metal stamps. Metal stamps shall not be used directly on cylindrical walls of the engine (except as prescribed on applicable drawings), but shall be placed on parts having high margins of safety.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Quality conformance inspection (see 4.7).

4.3 Qualification inspection.

4.3.1 Material. Samples of all materials used in the units shall be taken regularly for process and quality control and tested in accordance with a procedure proposed by the manufacturer and approved by the customer representative in order to insure uniformity and reliability of product. All ferromagnetic steel parts shall be subjected to magnetic particle inspection in accordance with MIL-I-6868 and MIL-STD-410. The equipment used shall be in accordance with MIL-I-6867. All non-destructive testing of materials and parts shall be in accordance with military specification MIL-I-6870. Hardness testing and tensile testing shall be in accordance with Federal Test Method 151.

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4.3.2 Test equipment. Performance parameters shall be determined by suitable test equipment to measure accurately the values required and shall be calibrated regularly to maintain the original accuracy of the test equipment. The calibration shall be under the cognizance of the customer representative. Equipment of the following types are included:

Thrust measuring
Scales
Pressure measuring
Recording
Miscellaneous

4.4 Testing propellant.

4.4.1 In process examination. Examination of the grains shall be in such a manner as to insure consistent compliance with quality control procedure.

4.4.2 Batch testing of propellant. Propellant grains acceptable on the basis of in-process inspection and examination per Paragraph 4.4.1 shall be batch tested in accordance with the conditioning and testing of rocket engines in Paragraph 4.4. Firing tests shall be in accordance with OP-5 for testing at a Government facility and DoD Manual 4145.26M if tested at a contractor's facility.

4.5 Igniter.

4.5.1 Mechanical inspection. Each igniter shall be inspected to insure that all parts are properly assembled and sealed in accordance with drawing requirements.

4.5.2 Electrical inspection. Each igniter assembled shall be inspected to insure that continuity of the electrical circuit exists and that the resistance of the igniter assembly components and leads is within the limits indicated on applicable drawings.

4.5.3 Ballistic evaluation. Each lot of igniters shall be sampled and tested to insure that the igniter assemblies of that lot meet the standards established for ballistic performance governing this rocket engine.

4.6 Rocket engines. Sampling and testing of rocket engines shall be made according to the following sampling systems as specified in the procurement contract to recognize three general levels of production quality controls.

4.6.1 Initial production schedule. Two (2) assembled rocket engines shall be selected at random representing each batch of propellant and tested by suitable equipment (4.3.2) to assure performance (3.1) under environmental conditions (3.7).

4.6.1.1 Testing at the temperature extremes. The testing system in this initial production schedule shall include performance measurement at both extreme temperatures of the firing temperature range (3.1.1) for each batch of propellant.

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4.6.1.2 Acceptance by revision of nozzle throat size. When the performance limits (3.1) are not satisfied in this testing but it appears that the performance can be brought within specification limits by revision of the nozzle throat size (3.4.3), the batch shall be accepted by the firing of one (1) additional engine, containing the revised nozzle throat size for the batch.

4.7 Quality conformance inspection. Once the rocket engine has been qualified under the first article inspection procedure, the inspection level may be reduced, providing that adequate quality control is maintained. Quality conformance may be used for an intermediate production schedule or a normal production schedule. For either schedule the acceptance criteria is still mandatory.

4.8 Intermediate production schedule.

4.8.1 Reduction to single sample from batch. When a satisfactory production level has been established, the sampling shall be reduced to one (1) assembled rocket engine representing each batch of propellant tested on the suitable equipment (4.3.2) to assure performance (3.1) under environmental conditions (3.7).

4.8.2 Alteration of environmental conditions. The testing at temperature extremes (3.1.1) shall be alternated in the single firing from sequential batches under the sampling plan.

4.8.3 Batch acceptance. Provision for batch acceptance by revision of nozzle throat size per Paragraph 4.6.1.2 applies hereunder.

4.8.4 Random selection of engines. Where stipulated in the production control system of the model specification, random selection of assembled engines (4.1.2) shall be replaceable by random selection of propellant grain(s) for assembly and testing prior to production assembly of the engines representing the entire batch.

4.9 Normal production schedule.

4.9.1 Statistical control charts. Shall be established for each of the performance parameters specified as conditions of acceptance (Paragraph 4.7). Additional charts shall be employed at the discretion of the manufacturer to assure continuance of a uniform and reliable product.

4.9.2 Random selection of engines. Sampling shall consist of one (1) engine selected at random from each of four (4) propellant batches selected at random (total of four engines) from production lot.

4.9.3 Production lot size. Size of the production lot shall be stated in the procurement contract.

4.9.4 Alteration of environmental conditions. The testing at temperature extremes (3.1.1) shall be made the same for each group of four engines representing a lot, but shall be alternated between the extremes for the testing of sequential lots.

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4.9.5 Definition of statistical group. The size of statistical groups, periodicity of control limits, and definition of statistical control index ratings shall represent proper statistical control practices and shall be cited in the applicable model specification.

4.9.6 Lot acceptance. Acceptance of the lot shall be based on satisfaction of the ballistic performance requirements (4.6) as determined by the control limits of the quality control charts. A lot not qualified under this section shall be examined in accordance with proper statistical resampling plans in a manner stipulated in the procurement contract.

4.10 Acceptance criteria. The acceptance of this rocket engine shall be based on satisfaction of the following performance requirements;

(a) Duration of thrust (3.1.2) in accordance with Figure 1.

(b) Average thrust (3.1.3) in accordance with Figure 2.

(c) Ignition interval (3.1.5) not to exceed 0.5 seconds at any temperature in the temperature firing range (3.1.1).

4.11 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirement of MIL-S-28786.

5. PACKAGING

5.1 Unit identification. The rocket engine shall have identification data as stated in Paragraph 3.9.

5.2 Unit packaging. The rocket engine shall be packaged per MIL-P-116, Method III. The igniter assembly shall be packaged per MIL-P-116, Method II.

5.3 Non-propulsive condition. The rocket engine shall be in a non-propulsive condition as prepared for shipment.

5.4 Hazard identification for fire fighting. During storage or transportation, appropriate hazard identification will be provided in accordance with DoD 5154.4S, Chapter 11.

5.5 Quantity-distance criteria. The quantity-distance criteria for the explosive materiel shall be in accordance with DoD 5154.4S.

6. NOTES

6.1 Intended use. These rocket engines are designed to provide sufficient thrust to an aircraft at take-off so that the take-off distance is substantially reduced, or minimized.

6.2 Definitions.

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6.2.1 Drawings. All drawings shall be defined in accordance with DoD-STD-100.

6.2.2 Quality assurance. All quality assurance terms shall be defined in accordance with MIL-STD-109.

6.2.3 Fuze explosive components. Fuze explosive components shall be defined in accordance with MIL-STD-320.

6.2.4 Jet thrust units. Nomenclature of jet thrust units shall be as defined herein (see 3.1.2, 3.1.3, 3.1.4).

6.3 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

6.4 Identification of changes. This specification has been revised to conform with MIL-STD-961 and the changes are not marked.

Preparing Activity
Navy - AS
(Project No. 2845-N012)

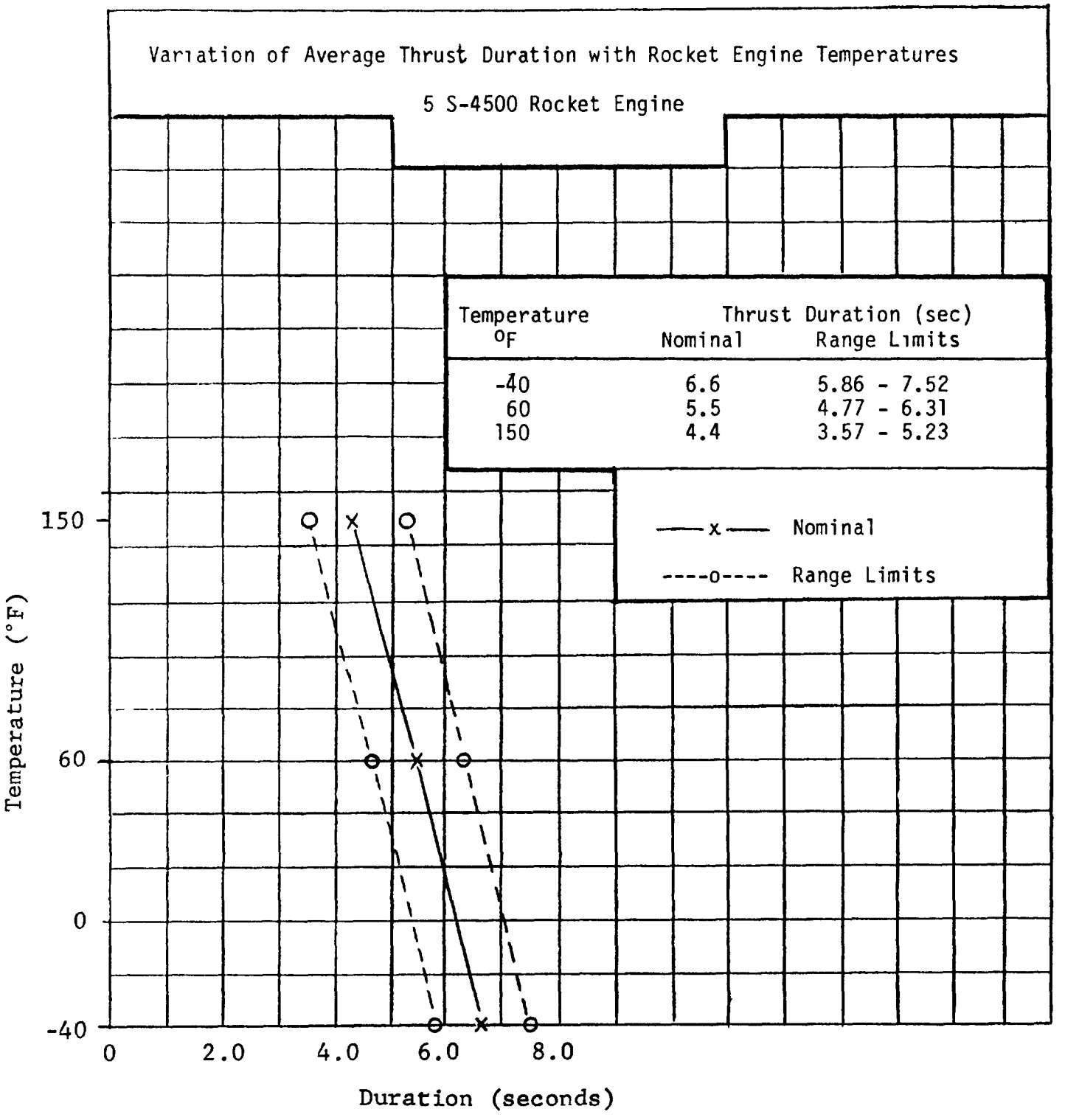
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Figure 1

Variation of Average Thrust Duration with Rocket Engine Temperatures

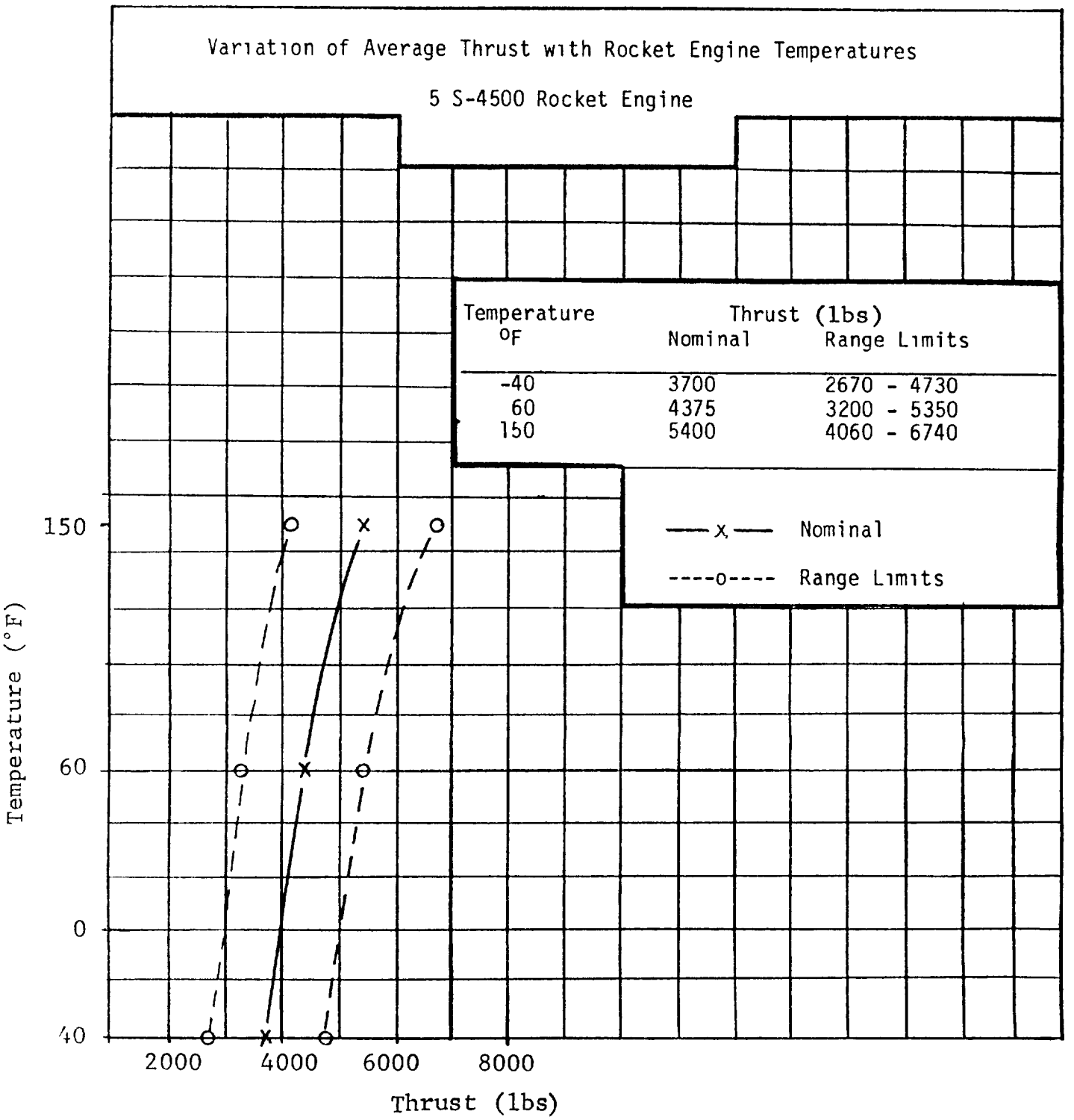
5 S-4500 Rocket Engine

Temperature °F	Thrust Duration (sec) Nominal	Range Limits
-40	6.6	5.86 - 7.52
60	5.5	4.77 - 6.31
150	4.4	3.57 - 5.23



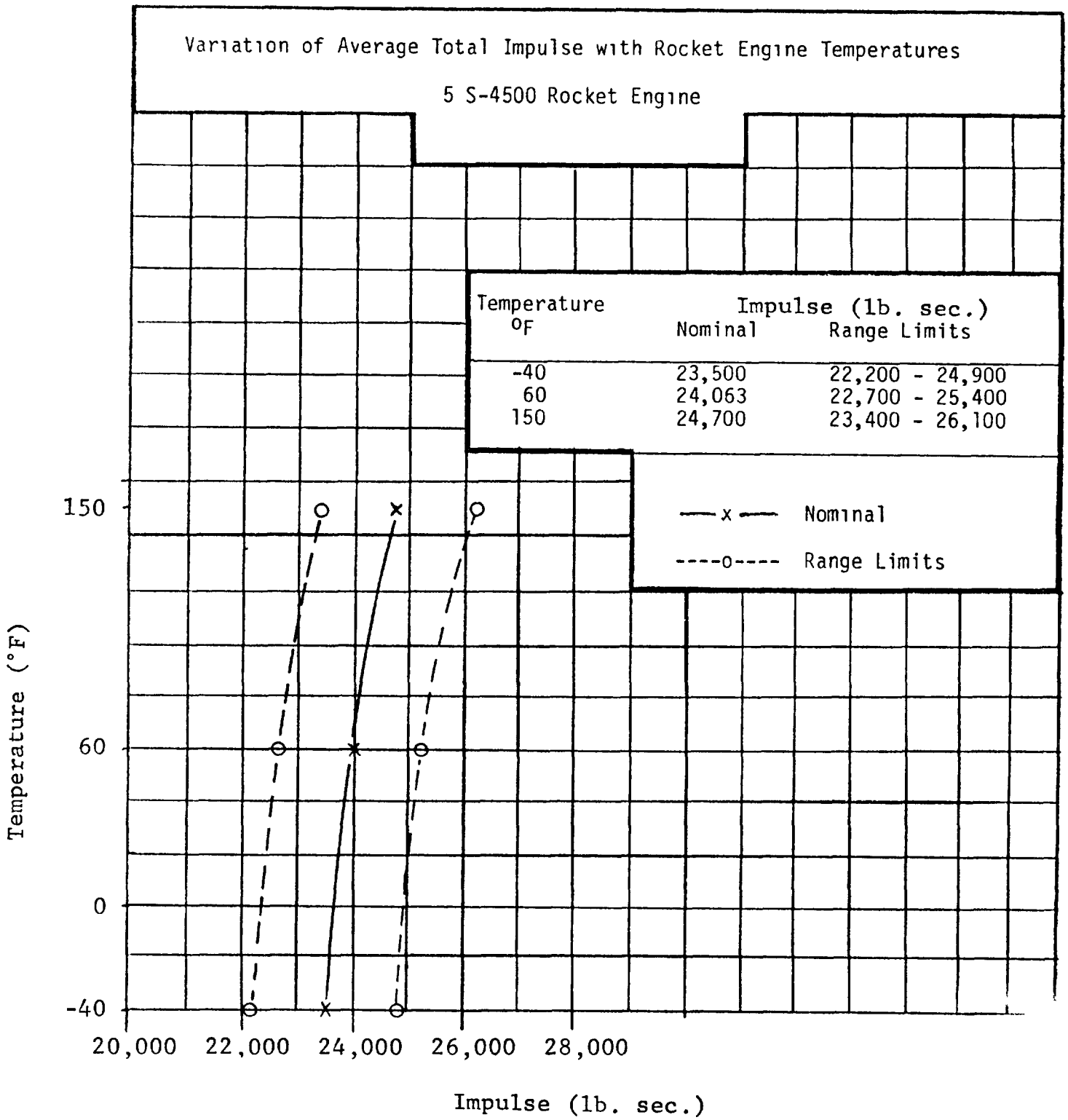
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Figure 2



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Figure 3



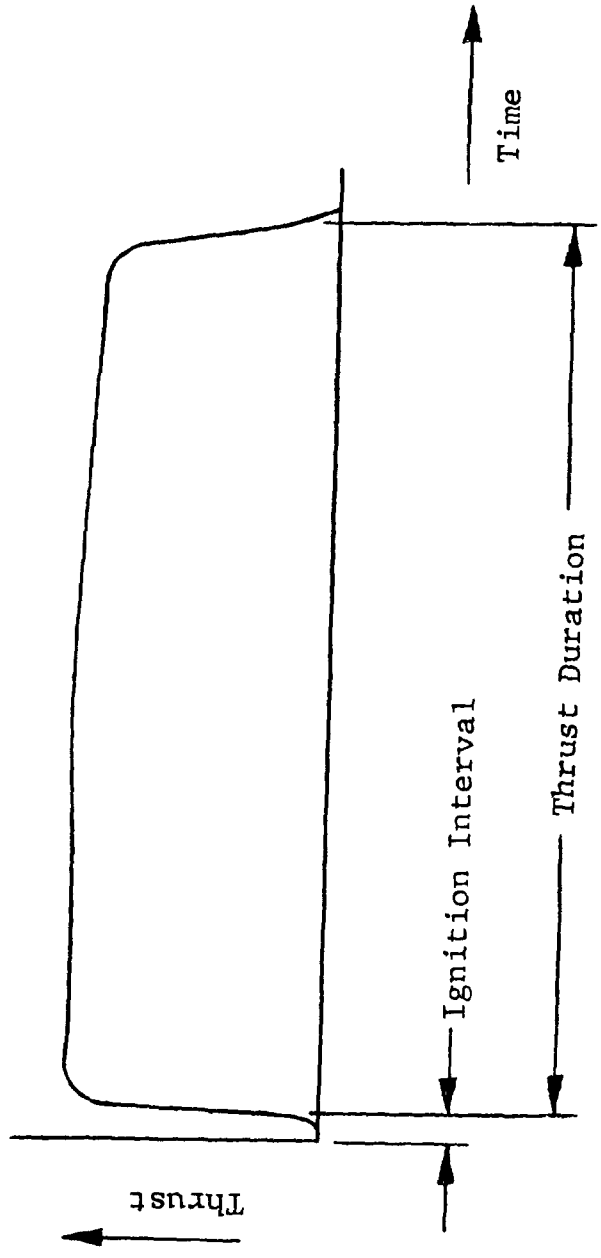
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Figure 4

5 S-4500

Sample Thrust-Time Curve

(Origin of the axis represents time of application of specified electrical impulse to igniter terminals)



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-E-8581A(AS)	Engine; Rocket, Solid Propellant 5 Second Duration, 4500 Pounds Thrust

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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